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ORIGINAL INVESTIGATION

The Association Between Alterations in Redox Homeostasis, Cortisol, and Commonly Used Objective and Subjective Markers of Fatigue in American Collegiate Football

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Purpose: To assess associations between a free oxygen radical test (FORT), free oxygen radical defense test (FORD), oxidative stress index, urinary cortisol, countermovement jump (CMJ), and subjective wellness in American college football. ***Methods:*** Twenty-three male student athlete American college football players were assessed over 10 weeks: off-season conditioning (3 wk), preseason camp (4 wk), and in season (3 wk). Assessments included a once-weekly FORT and FORD blood sample, urinary cortisol sample, CMJ assessment including flight time, reactive strength index modified and concentric impulse, and a daily subjective wellness questionnaire. Linear mixed models analyzed the effect of a 2 within-subject SD change in the predictor variable on the dependent variable. The effects were interpreted using magnitude-based inference and are presented as standardized effect size (ES) ± 90% confidence intervals. ***Results:*** *Small* negative associations were observed between FORT–flight time, FORT–fatigue, FORT–soreness (ES range = –0.30 to –0.48), FORD–sleep (ES = 0.42 ± 0.29), and oxidative stress index soreness (ES = 0.56 ± 0.29). *Small* positive associations were observed between FORT–cortisol (ES = 0.36 ± 0.35), FORD–flight time, FORD reactive strength index modified and FORD–soreness (0.37–0.41), oxidative stress index concentric impulse (ES = 0.37 ± 0.28), and with soreness–concentric impulse, soreness–flight time, and soreness reactive strength index modified (0.33–0.59). *Moderate* positive associations were observed between cortisol–concentric impulse and cortisol–sleep (0.57–0.60). ***Conclusion:*** FORT/FORD was associated with CMJ variables and subjective wellness. Greater amounts of subjective soreness were associated with decreased CMJ performance, increased FORT and cortisol, and decreased FORD.

Keywords: FORT/FORD, oxidative stress, countermovement jump, subjective wellness questionnaire, monitoring

American collegiate football (ACF) competition typically comprises 12 games with the potential for postseason conference championship and bowl games; it is, thus, paramount that players be available to play as the season is too short for a diminished roster. During the off season, student athletes are exposed to demanding physical workloads incorporating heavy weightlifting alongside on-field conditioning coupled with an intense "camp phase" comprising football-specific and scrimmage-based sessions.[1] Camp phases represent a 2½-fold increase in workload relative to the off season due to the change in allotted hours for practice and football commitments moving from 8 to 20 hours per week. Compounding the physical workload, the added stress of academic commitments can also increase the risk of injury and illness in this population.[2] Therefore, regimented monitoring systems are required to assess the response of the student athlete.

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Wellness questionnaires are used to subjectively assess the physiological stress associated with physical activity in ACF with soreness, sleep, and energy shown to be decreased up to 4 days postgame.[3] However, subjective measures can be associated with a high number of false positive outcomes wherein associations are observed as a simple consequence of a high reporting frequency.[4] Alternative measures include performance tests, such as the countermovement jump (CMJ), which has been shown to be a reliable tool in team sports to assess neuromuscular fatigue.[5,6] In ACF, CMJ performance has been shown to be reduced at half time before returning to pregame values almost immediately after the game[7]; however, CMJ performance across an entire ACF season is not known. Objective monitoring tools can also be used to assess individual responses from prescribed workloads.[8] Cortisol, for example, is commonly measured in response to exercise in elite sport to assess anabolic and catabolic processes from physical activity,[9] which are known to be elevated due to muscle tissue damage after contact sporting events[7] and remain elevated for up to 96 hours following competitive contact sport match play.[5] Yet in ACF, cortisol concentrations appear similar pregame and postgame,[10] during the preseason, and throughout the competitive season despite considerable variation.[11]

Alternate objective biochemical assessments of the physiological mechanisms effected by ACF training/match play are, thus, desirable. The free oxygen radical test (FORT) and free oxygen radical defense test (FORD) is a reliable and repeatable point of care (POC) test assessing alterations in redox homeostasis.[12] It represents the balance between reactive nitrogen oxygen species (RNOS) and endogenous antioxidants to determine an individual's

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prooxidant state, quantified by the index of FORT and FORD to give the oxidative stress index (OSI).[13] Herein, prooxidant and antioxidants identified by FORT and FORD measures have been shown to acutely elevate in response to an exercise test with samples taken at submaximal levels and at maximal termination 20 minutes after a static recovery period in elite distance runners.[14] Furthermore, elevations are seen to be evident upward of 48 hours postmatch in elite soccer players,[15] and an association exists between training load and oxidative stress throughout a soccer season.[16] However, the impact of ACF training and competition on oxidative stress is not known. This investigation will, therefore, examine associations between oxidative stress and other commonly used measures of athlete readiness and fatigue throughout the ACF season.

Methods

Participants

In this study, 23 male student athlete ACF players composed of 10 defensive and 13 offensive (mean [SD]; 20.1 [1.4] y, 108 [20] kg, 187 [8] cm) players participating in the same division 1 A 2017 collegiate football team were assessed over a 10-week period. Experimental assessments were performed during off-season conditioning (3 wk), preseason camp (4 wk), and in season (3 wk). Ethical approval was gained from the University of Wollongong Ethics Committee and the University of Oregon's research compliance services. Written consent was obtained from each player prior to commencing this study.

Physical Training

This study captured 3 distinct phases of the 2017–2018 National Collegiate Athletic Association football season including: (1) "Off-season conditioning," (2) "Camp," and (3) "In season." Off-season conditioning comprised 4 strength sessions, 2 of which included general field-based conditioning each week, totaling 8 hours per week. Camp included 6 football practice sessions and 2 strength sessions per week, totaling <20 hours per week in physical activities. Each practice session was broken into 2 distinct parts, position-specific practice and game-style practice. Strength sessions during camp included lighter compound movements compared with off-season weight programs. In season, each week included 1 game day, 2 contact football practices, 2 low-intensity walk-throughs, and 1 practice/recovery session, totaling approximately 12 hours per week.

Experimental Design

Players were assessed on 1 day each week (Monday–Friday, randomly allocated) upon arrival at the training facility for 10 weeks, cumulating a maximum of 23 individual weekly data points.

Blood Samples. The FORT and FORD were analyzed from a fingertip blood sample, which was taken between 6.00 and 7.00 AM on sampling days. Participants arrived at the practice facility in a fasted state with the exception of consuming up to 500 mL of water ad libitum. Participants were seated, and the fingertip was cleansed with an alcohol swab and left to dry. The participant was then lanced at a depth of 1.6 mm, and the first drop of blood was wiped from the skin with a cotton bud to avoid contamination. About 300 μL of capillary blood was drawn into a heparinized capillary tube, which was capped, immediately refrigerated at 4°C, and analyzed within 30 minutes. A total of 50 and 20 μL of capillary blood were transferred into separate capillary tubes for FORT and FORD analysis, respectively. The appropriate reagents were added, inverted several times to mix, centrifuged at 5000 revolutions per minute (2000*g*) for 1 minute, and analyzed at 37°C with an absorbance wavelength of 505 nm using a Callegari CR3000 (Callegari SpA; Catellani Group, Parma, Italy) according to the manufacturer's instructions.[12] A total of 324 FORT and FORD samples were collected.

Urine Samples. Participants were instructed to urinate in a 3-L container during a period of 10 hours (approximately 7:00 PM–5.00 AM) with the last sample being the first urination upon awakening. The volume was measured to the nearest 5 mL to define the whole overnight urine sample, aliquoted into 5 mL vials, and frozen at −20°C. Urinary cortisol was measured in duplicate using a standard enzyme-linked immunosorbent assay kit (Abcam, Cambridge, UK) and plate reader to quantify absorbance with sample values interpolated from the standard curve. A total of 193 urine samples were collected.

Countermovement Jump. Immediately after blood samples were taken, 3 CMJs were performed on commercially available force platforms and analyzed using ForceDecks software (NMP Technologies, London, United Kingdom). Participants were instructed to stand on a dual force platform (AMTI BP-600-900, Advanced Mechanical Technology, Watertown, MA) with one foot on each, place hands on hips, and jump as high as possible. A single jump (best recorded flight time) was chosen for analysis with flight time calculated as the duration of time the athlete was off the force plate.[17] Reactive strength index modified (RSImod) is a reliable measure of an athlete's time spent on the ground generating force compared with the time that was spent in the air.[18] Finally, concentric impulse (in newton seconds) was calculated from the area under the force–time curve[19] during the concentric phase. A total of 241 CMJ tests were collected.

Subjective Wellness Questionnaires. Each morning, prior to training (6.00–7.00 AM), all players completed a customized subjective wellness questionnaire assessing perceived sleep quality, overall muscle soreness, and fatigue using a 5-point Likert scale wherein lower values indicated a negative response and higher values indicated a positive response.[3] A total of 2086 wellness questionnaires responses were collected.

Statistical Analysis

Linear mixed models (lme4 package in R; version 1.0.136)[20] were used to assess the association between biochemical markers (FORT, FORD, OSI, and cortisol), CMJ metrics, and subjective wellness questions. Furthermore, the associations of CMJ variables were also assessed with subjective wellness questions. The FORT, FORD, and OSI were set as the dependent variables with cortisol, CMJ metrics, and subjective wellness questions set as independent variables, totaling 21 models. Second, cortisol was set as the dependent variable with FORT, FORD, OSI, CMJ metrics, and subjective wellness questions set as the independent variables, totaling 9 models. Finally, associations between CMJ metrics and subjective wellness questions were examined with subjective wellness questions set as the dependent variable and CMJ metrics as the independent variables, totaling 9 models. In total, there were 39 models conducted in this analysis, which can be a limitation due to multiple testing. All models were set with a random intercept for the individual as a random effect. Each relationship was standardized by multiplying the final slope by 2 × within-subject SD, using a mixed-model reliability analysis

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method.[21] This method was chosen to reflect the effect of a 2 SD change in the predictor variable (ie, the change needed to go from a typically low to a typically high subject) on the dependent variable. This change was then converted to an effect size and 90% confidence intervals using the between-subject SDs that were categorized as <0.2, trivial; 0.21 to 0.60, small; 0.61 to 1.20, moderate; 1.21 to 2.0, large; and >2.0 very large. Effect sizes were considered real when greater than 0.20 and were described as 75% to <95%, *likely*; 95% to <99.5%, *very likely*; >99.5%, *most likely*.[21]

Results

Small negative associations were observed between FORT–flight time, FORT–fatigue, and FORT–soreness (Figure 1A). *Small* positive associations were observed between FORD–flight time,

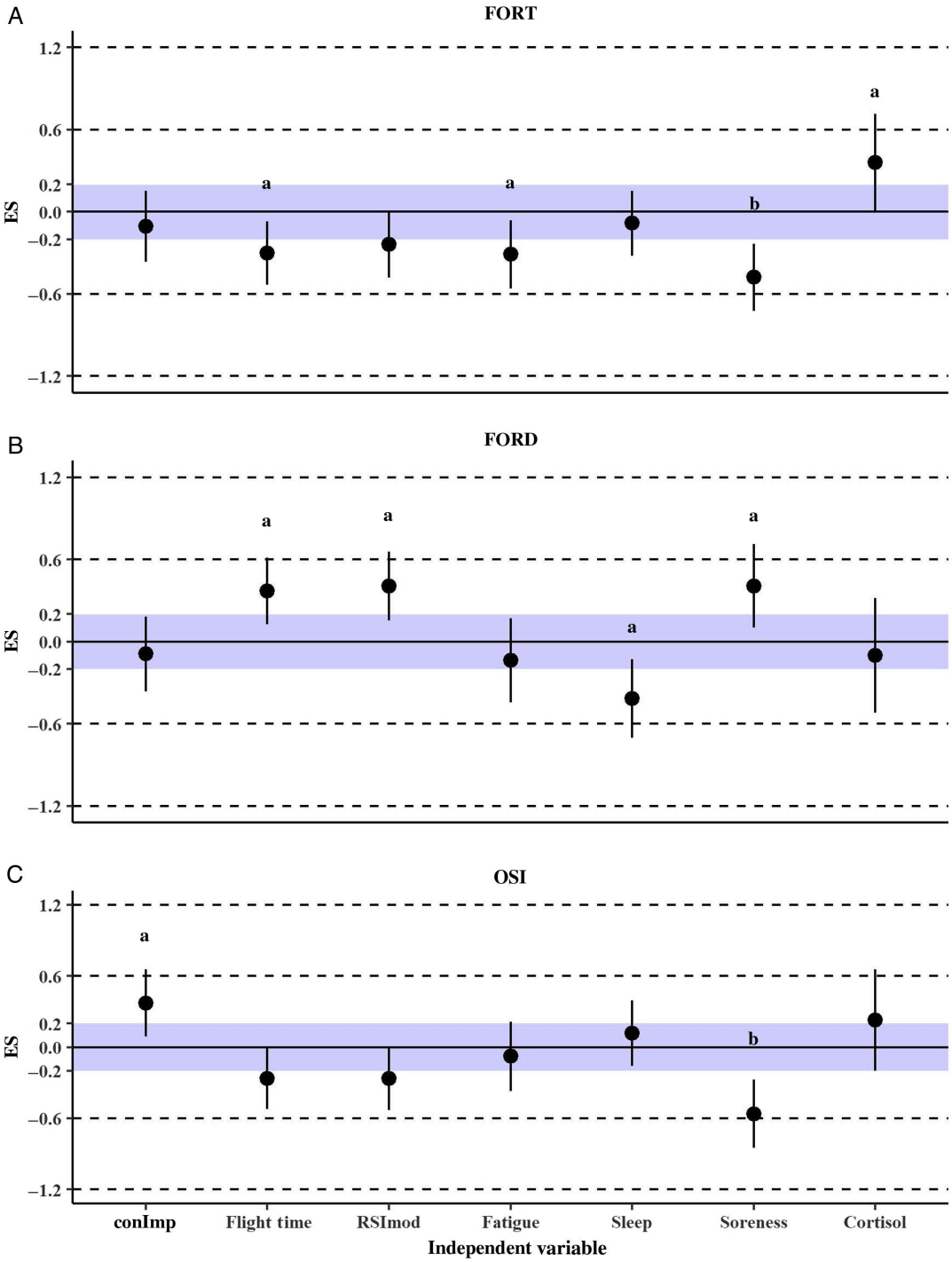

Figure 1 — Associations between FORT (A), FORD (B), and OSI (C) with countermovement-jump variables (conImp and RSImod) and subjective reports of sleep, soreness, and fatigue. conImp indicates concentric impulse; ES, effect size; FORD, free oxygen radical defense test; FORT, free oxygen radical test; OSI, oxidative stress index; RSImod, reactive strength index modified. [a] Effect *likely* >0.2. [b] Effect *very likely* >0.2.

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FORD–RSImod, and FORD–soreness, and a *small* negative association was observed between FORD–sleep (Figure 1B). *Small* positive associations were observed between OSI–concentric impulse, and a *small* negative association was observed with OSI–soreness (Figure 1C).

Moderate positive associations were observed between cortisol–concentric impulse and sleep (Figure 2).

Small positive associations were observed between soreness–concentric impulse, soreness–flight time, and soreness–RSImod (Figure 3).

Discussion

In this investigation, FORT, FORD, and OSI appeared to offer insights into the prooxidant status in ACF with fluctuations consistently associated with subjective reports of muscle soreness and objective assessments of CMJ and cortisol throughout the ACF season. Improved CMJ performance was also observed when the group reported lower levels of subjective muscle soreness. However, although cortisol was positively associated with FORT, no other associations were observed with the objective and subjective measures of stress examined within this investigation.

A decrease in predictor variables, subjective soreness and fatigue, and flight time, variables shown to be negatively affected by increased game and training volumes,[3,5] had a negative effect on FORT, highlighting greater circulating concentrations of RNOS. The association between cortisol and FORT observed in this investigation aligned with previous research as both have been shown to respond to physical activity.[5,14] Furthermore, the association may be reflective of the demands of ACF practice and games that involve explosive movements, such as accelerating, decelerating, and cutting,[1] which along with muscle contusions from contacts, result in increased muscle microtrauma.[22] Subsequently, the stress of ACF games is associated with increases in subjective muscle soreness.[3] Existing research has shown increases in the prooxidant response alongside delayed-onset muscle soreness and decreases in vertical jump performance 24 hours following soccer games.[23] The findings in this present study, thus, similarly suggest that increasing concentrations of FORT align with increased muscle damage, decreased CMJ, and increased muscle soreness associated with the demands of contact sports.[24]

In contrast to the negative associations observed relative to increasing concentrations of FORT, the countermeasure FORD displayed positive associations with flight time and RSImod and was elevated alongside reduced levels of subjective muscle soreness. Endogenous antioxidants concentrations have previously been shown to acutely increase postmaximal exercise[14] and remain elevated up to 48 hours postsoccer matches.[23] Furthermore, after intense periods of training, circulating endogenous antioxidants are also chronically elevated, representing a mechanism to cope with the increased concentrations of RNOS observed following heavy periods of training.[13,23] Subsequently, the reduction in subjective sleep quality observed alongside increased FORD in this investigation was likely associated with increased training volumes. It is well known that sleep is negatively affected postcompetition[25] as well as during intensified periods of training in a similar cohort of

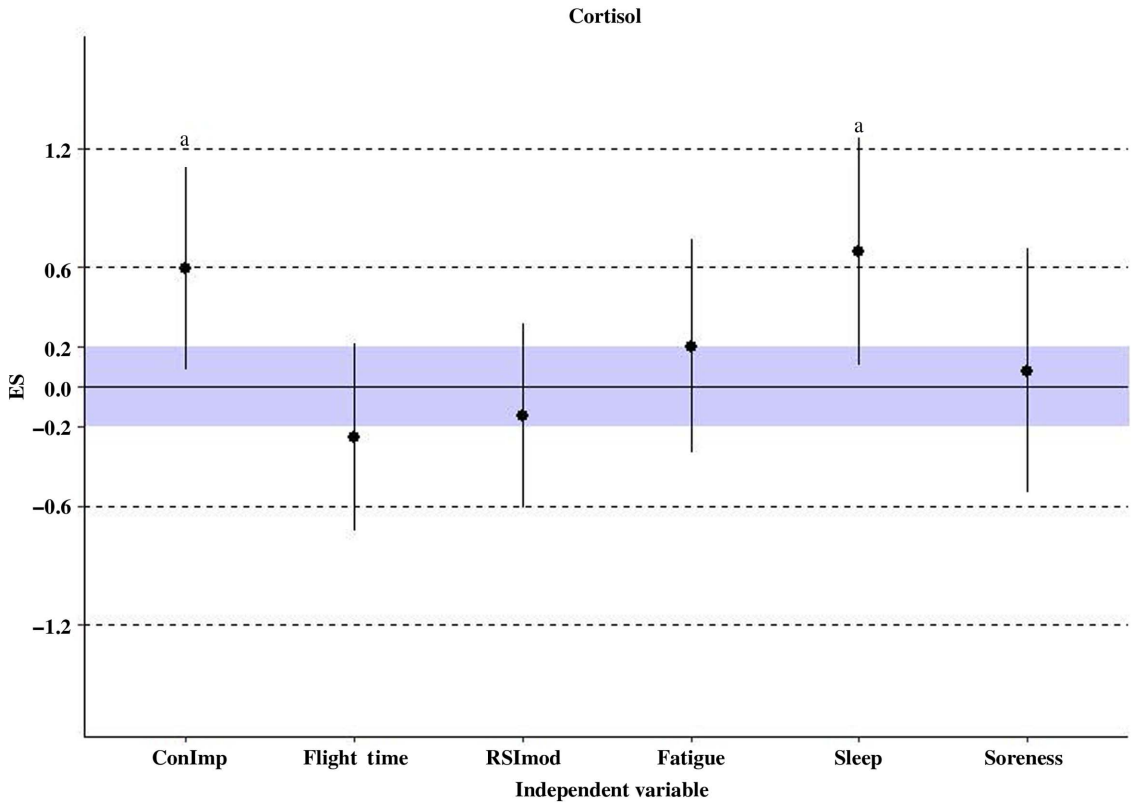

Figure 2 — Associations between cortisol with FORT, FORD, OSI, and countermovement-jump varibles (conImp and RSImod) and subjective reports of sleep, soreness, and fatigue. conImp indicates concentric impulse; ES, effect size; FORD, free oxygen radical defense test; FORT, free oxygen radical test; OSI, oxidative stress index; RSImod, reactive strength index modified. [a] Effect *likely* >0.2.

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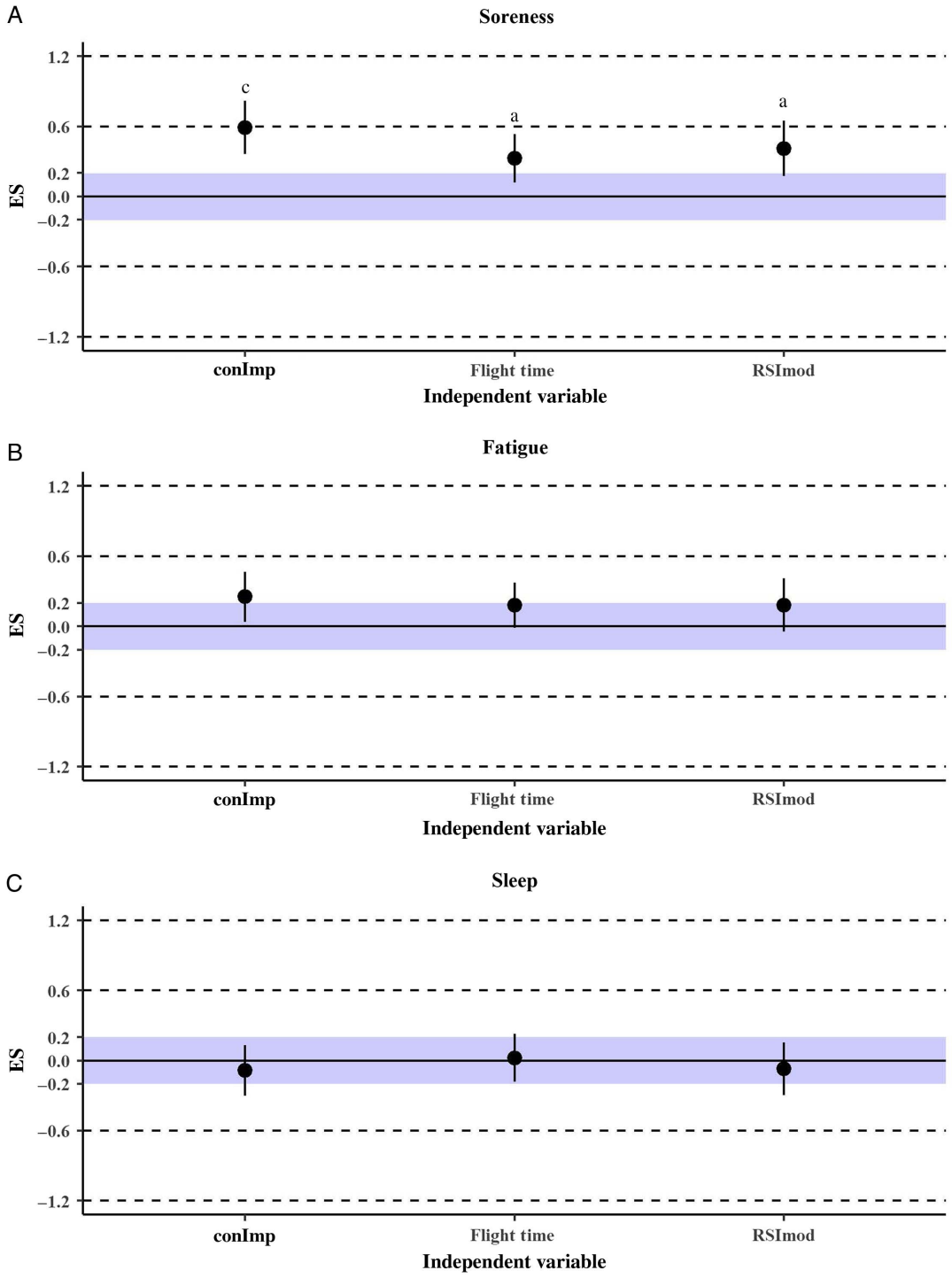

Figure 3 — Associations between subjective reports of sleep, soreness, and fatigue and countermovement-jump variables (conImp and RSImod). conImp indicates concentric impulse; ES, effect size; RSImod, reactive strength index modified. [a] Effect *likely* >0.2. [c] Effect *most likely* >0.2.

ACF players.[26] Therefore, it may be assumed that the increases in flight time and RSImod and decreased subjective muscle soreness found may be a product of appropriate training periodization and recovery. The FORD was associated with more measures than FORT in this investigation. As such, independently, FORD may give insight into the demand of endogenous antioxidants from increased levels of RNOS caused by the stress of exercise of lifestyle habits. The index of FORT and FORD, OSI, did not have any significant associations other than with subjective muscle soreness. As this measure is an index of both FORT and FORD, results may be augmented when analyzed as an index. Furthermore, an unexpected association emerged; an increase in concentric impulse was found to be associated with increased OSI. This observation is surprising given that RNOS are known to decrease

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the contractile ability of the muscle[27] and that no other CMJ variable was associated with a changed OSI. The consistency observed in the associations between subjective muscle soreness and redox homeostasis and the increased strength with the ratio of FORT and FORD suggests that this combination of measures may be useful in the understanding of the physiological response to athletic commitments.

Indeed, perceived soreness was not only consistently associated with changes in FORT, FORD, and OSI but also with all measured CMJ variables. The associations observed between CMJ performance and muscle soreness is a finding that may be worth further investigations in ACF around the response to training and games loads. When comparing the positive associations observed in this investigation, indicative of increased performance alongside lower levels of muscle soreness, previous research in a rugby league found increased subjective muscle soreness alongside decreased CMJ flight time following match play.[24] Both outcomes favorably represent the usefulness of perceived soreness as a perceived measure of wellness, recovery, and athlete readiness. This is also meaningful given the frequency and logistical ease with which subjective questionnaires can be completed and their association with changes in training and game loads.[3,28] This may confirm that subjective wellness questionnaires including overall muscle soreness may be a best fit as a standard longitudinal assessment. Although it is well known that there are considerations around unvalidated questionnaires, high false positives, and questionnaire fatigue,[4,29] the FORT/FORD test appears to be one that could highlight whether athletes reporting increased levels of subjective soreness are also in a prooxidant state known to be detrimental to recovery and adaptation.[13] However, subjective muscle soreness did not appear to be responsive to changing cortisol concentrations in this investigation.

Other than the expected association with FORT, cortisol was not associated with any of the other variables measured in this investigation. Cortisol has, to date, been used as one of the more common biochemical markers of stress in sport, typically shown to fluctuate in response to acute exercise.[5,30] It must be considered that within this investigation cortisol was sampled via urine samples, whereas the majority of the listed investigations used blood sampling. In opposition to the findings of the present study, research within Australian Rules Football reported increased flight time: contraction time alongside elevated cortisol postmatch.[5] Another study in ACF found pre–post game cortisol concentrations to appear stable in the playing group; however, significant decreases were observed in the nonplaying group in the 42- to 44-hour period postmatch.[10] However, although on face value this may suggest that cortisol is not a reliable marker of ACF stress, these results may have been influenced by the precompetition rise in cortisol that is commonly seen in contact sports.[31] Furthermore, a study that assessed the response of cortisol across a season found that there were no significant changes across time points, although it did see large variations.[11] Furthermore, the positive association between cortisol and subjective sleep quality we observed was also unexpected. It is understood that when athletes are experiencing elevated workloads, in turn increasing concentrations of cortisol,[6] sleep is negatively affected.[25] As cortisol is responsive to a range of stressors from environmental, physical, and psychological, this measure may not to be suitable to investigate the response of physical activity in ACF. The associations that have been found within this study may suggest that the response of FORT and FORD measured via a novel POC test aligns with commonly used measures to highlight various forms of fatigue and may be a measure used within a team sport environment to highlight the prooxidant status of ACF student athletes. However, one should consider that samples were randomized across the training week, which may not have accounted for the acute response to daily variation in the training stimuli (eg, heavy vs light training). Further investigations assessing FORT/FORD across training days in a single a training week and across the different phases of the ACF season, thus, appear warranted. The consistent associations found with subjective soreness in this study may, however, allow practitioners to explore the implementation of subjective wellness questionnaires within their athlete response monitoring protocols alongside CMJ testing.

Practical Implications

- Subjective muscle soreness seems to be a valid indicator of stress within ACF players and, therefore, should be implemented within athlete monitoring systems.
- The POC biochemical testing presents as a logistically viable method of assessing alterations in redox homeostasis among ACF players.
- The FORT/FORD test may be used to confirm changes found in routine subjective wellness questionnaire and CMJ test results.
- The FORT/FORD test has associated cost implications; subjective muscle soreness may be a cost-effective method to infer potential oxidative stress found in ACF players, although due to the effect size of these associations, further investigation may be required.
- Urinary cortisol was not found to be useful as a marker of physiological stress within this investigation.

Conclusion

This study investigated the associations between prooxidant and antioxidant status via the FORT/FORD test, a novel POC test, with subjective wellness questionnaire questions and a CMJ assessment. The FORT and FORD were both found to have associations with subjective wellness questions and CMJ variables. Associations found highlight the potential implementation of an objective POC test assessing prooxidant and antioxidant status as a proxy of oxidative stress in ACF players. Subjective muscle soreness was found to be associated with decreased CMJ performance, increased FORT and cortisol, and decreased FORD. Therefore, assessing subjective muscle soreness on a routine basis may be a time-efficient, noninvasive response measure.

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